SCSE 2123 SOFTWARE REQUIREMENTS ENGINEERING

CHAPTER 3:Practices for Requirements Elicitation Part 2

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1

Contents

- Gathering Techniques
- Design and Idea-Generating Techniques

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2

Objectives

By the end of this topic, students will be able to:

i. Differentiate and justify the use of various elicitation techniques (e.g., interview vs. workshop vs. crowd-based RE) based on project context and stakeholder characteristics

ii. Critically assess the effectiveness and limitations of chosen techniques for specific project requirements

iii. Design and propose an integrated elicitation plan combining gathering, design, and idea-generating techniques to address a real or simulated software project

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3

Requirements Elicitation Techniques

Kano Model

Gathering Techniques

- Questioning
- Observation
- Collaboration
- Artifact-based

Design and Idea-generating Techniques

- Creativity
- Design

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4

3.3 REQUIREMENTS ELICITATION TECHNIQUES:
- GATHERING TECHNIQUES

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Gathering Techniques

Mainly useful to elicit performance and basic requirements by investigating different sources

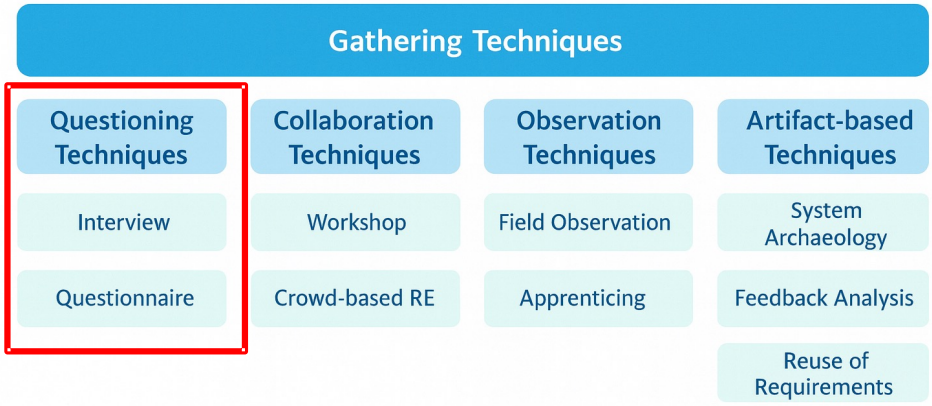
Gathering Techniques			
Questioning Techniques	Collaboration Techniques	Observation Techniques	Artifact-based Techniques
Interview	Workshop	Field Observation	System Archaeology
Questionnaire	Crowd-based RE	Apprenticing	Feedback Analysis
			Reuse of Requirements

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6

Gathering Techniques

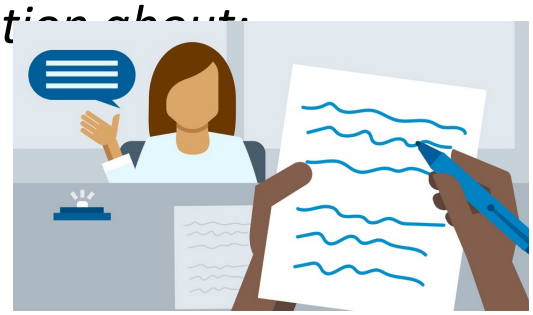


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7

INTERVIEWING

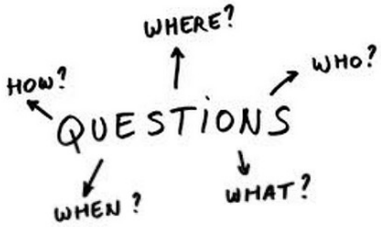
- *Interviewing is an important method for collecting data on human and system information requirements*
- *Interviews reveal information about:*
  - *Interviewee opinions*
  - *Interviewee feelings*
  - *Goals*
  - *Key HCI concerns*



8

INTERVIEW PREPARATION

- *Reading background material*
- *Establishing interview objectives*
- *Deciding whom to interview*
- *Preparing the interviewee*
- *Deciding on question types and structure*





9

2 MAIN TYPE OF QUESTIONS

OPEN-ENDED QUESTIONS

- *Open-ended interview questions allow interviewees to respond how they wish, and to what length they wish*
- *Open-ended interview questions are appropriate when the analyst is interested in breadth and depth of reply*
- *Example:*
  1. *What is your opinion of current e-Learning system, how to improve it?*
  2. *Once a user is registered on your website, what are the next steps/processes?*

CLOSED QUESTIONS

- *Closed interview questions limit the number of possible responses*
- *Closed interview questions are appropriate for generating precise, reliable data that is easy to analyze*
- *The methodology is efficient, and it requires little skill for interviewers to administer*
- *Example:*
  1. *How many users are registered in your system?*
  2. *Who receives the online customer's complaint?*

10

OPEN-ENDED QUESTIONS ADVANTAGES & DISADVANTAGES



ADVANTAGES

- Puts the interviewee at ease
- Allows the interviewer to pick up on the interviewee's vocabulary
- Provides richness of detail
- Reveals avenues of further questioning that may have gone untapped
- Provides more interest for the interviewee
- Allows more freedom
- Makes phrasing easier for the interviewer
- Useful if the interviewer is unprepared



DISADVANTAGES

- May result in too much irrelevant detail
- Possibly losing control of the interview
- May take too much time for the amount of useful information gained
- Potentially seeming that the interviewer is unprepared
- Possibly giving the impression that the interviewer is on a "fishing journey"

11

CLOSED QUESTIONS ADVANTAGES & DISADVANTAGES



ADVANTAGES

- Saving interview time
- Easily comparing interviews
- Getting to the point
- Keeping control of the interview
- Covering a large area quickly
- Getting to relevant data



DISADVANTAGES

- Boring for the interviewee
- Failure to obtain rich detailing
- Missing main ideas
- Failing to build rapport between interviewer and interviewee

12

OTHER TYPES OF QUESTIONS

BIPOLAR QUESTIONS

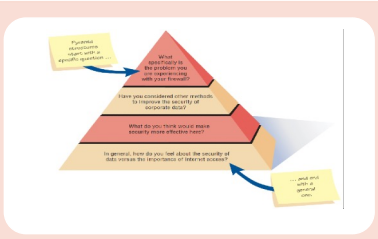
- *A special kind of closed question*
- *Bipolar questions are those that may be answered with a "yes" or "no" or "agree" or "disagree"*
- *Bipolar questions should be used carefully*
- *Probing questions elicit more detail about previous questions*

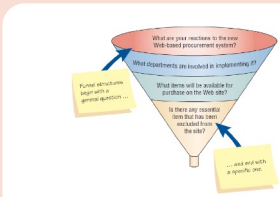
PROBE QUESTIONS

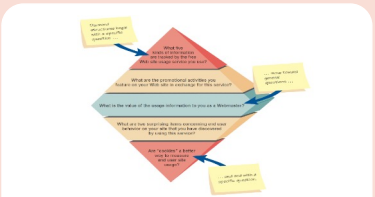
- *The purpose of probing questions is:*
  - *To get more meaning*
  - *To clarify*
  - *To draw out and expand on the interviewee's point*
- *May be either open-ended or closed*
- *E.g., "Why?" or "Please explain more for me"*

13

ARRANGING QUESTIONS







PYRAMID

- *Starting with closed questions and working toward open-ended questions*

FUNNEL

- *Starting with open-ended questions and working toward closed questions*

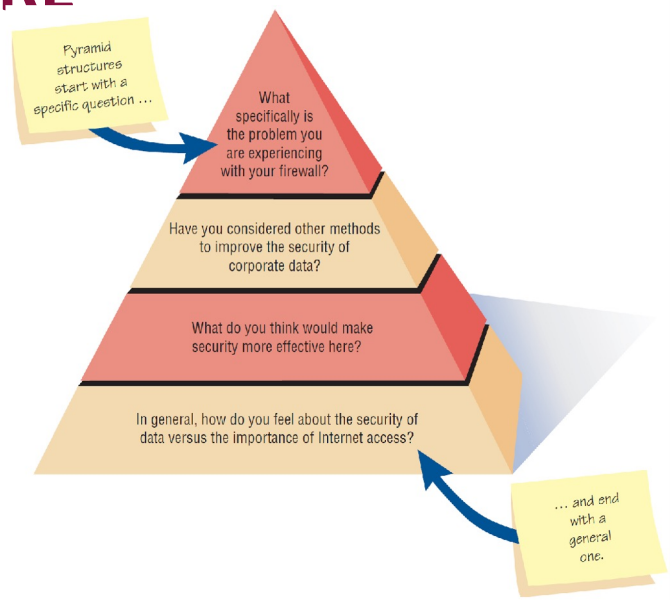
DIAMOND

- *Starting with closed, moving toward open-ended, and ending with closed questions*

14

PYRAMID STRUCTURE

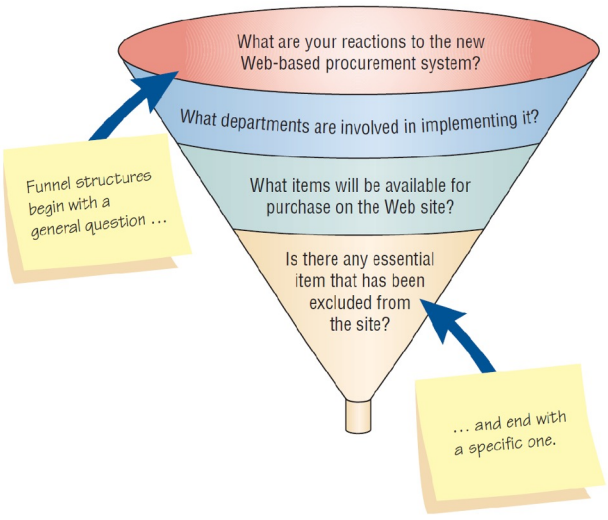
- *Begins with very detailed, often closed questions (from Specific to General Questions)*
- *Expands by allowing open-ended questions and more generalized responses*
- *Is useful if interviewees need to be warmed up to the topic or seem reluctant to address the topic*



15

FUNNEL STRUCTURE

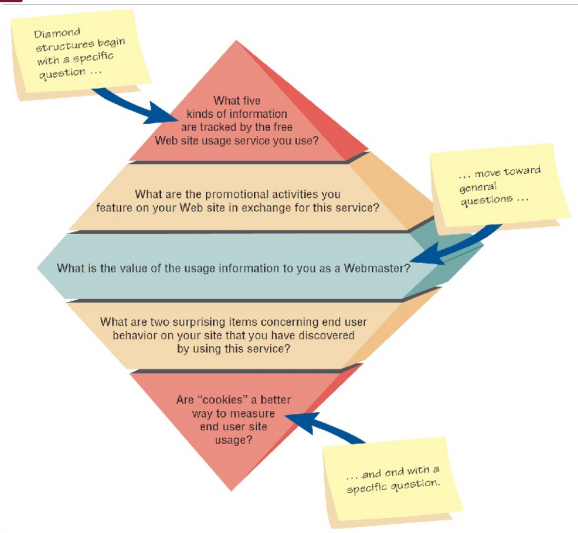
- *Begins with generalized, open-ended questions (from broad questions then funnels to specific questions)*
- *Concludes by narrowing the possible responses using closed questions*
- *Provides an easy, nonthreatening way to begin an interview*
- *Is useful when the interviewee feels emotionally about the topic*



16

DIAMOND STRUCTURE

- *A diamond-shaped structure begins in a very specific way*
- *Then more general issues are examined*
- *Concludes with specific questions*
- *Combines the strength of both the pyramid and funnel structures*
- *Takes longer than the other structures*



17

CLOSING THE INTERVIEW

- *Always ask "Is there anything else that you would like to add?"*
- *Summarize and provide feedback on your impressions*
- *Ask whom you should talk with next*
- *Set up any future appointments*
- *Thank them for their time and shake hands.*

18

INTERVIEW REPORT

- *Write as soon as possible after the interview*
- *Provide an initial summary, then more detail*
- *Review the report with the respondent*



19

Interview Techniques

- Open Questions
  - How? When? Where? How often? Who? Why? Which? What are the reasons?
- Listen
  - Raise a question, listen, wait (3s), listen, wait (3s) – next question
- Yesterday
  - Work on a typical day of your stakeholder, step by step
- Dive into / Drill down
  - "Do you use X ...", "For what do you use X...", "How exactly do you use X..."
- Filter/Closed Questions
  - "Do you know Z"?, "Try to explain Z.", "Have you ever used Z as follows...?"
  - "Do you use X or Y?"
- Summarize (Validation)
  - Summarize the recorded answers in your own words
- Concern
  - Tell me what really hurts, hinders you at your work
- Magic Question
  - "Imagine a fairy appears and will fulfill you three wishes for the new system. What would you wish for?"

20

Interview

- *Helps capture both high-level and detailed requirements.*
- *Typically uncovers requirements classified as* **satisfiers** *(explicit knowledge from stakeholders).*
- *Can reveal* **dissatisfiers** *or* **delighters** *by applying the right questioning techniques and observing user reactions.*
- *Allows new information to be discussed and clarified immediately.*
- *Requires an experienced interviewer to ensure completeness and accuracy of responses.*
- *Can be* **time-intensive**, *both in conducting the interviews and analyzing results.*

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21

Questionnaire

- *A commonly preferred technique for collecting input from large groups.*
- *Involves asking a broader group of stakeholders to respond to the same set of questions (delivered orally, in writing, or online).*
- *Questions are structured and standardized for consistency.*
- *Designing an effective questionnaire requires significant effort and subject knowledge.*
- *Does not allow for immediate feedback or clarification of responses.*
- *Typically used as a follow-up after initial insights have been gathered through interviews.*
- *Useful for validating preliminary ideas and confirming them across a wider audience.*

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22

QUESTIONNAIRES

Questionnaires are useful in gathering information (using open-ended and/or closed) from key organization members about:

- *Attitudes of people*
- *Beliefs of people*
- *Behaviors*
- *Characteristics*



QUESTIONNAIRES

- *Planning for the use of questionnaires*
  1. *Choose words carefully, department or unit, managers or supervisors*
  2. *Organization members are widely dispersed*
  3. *Many members are involved with the project*
  4. *Exploratory work is needed*
  5. *Problem solving prior to interviews is necessary*
- *Questions are designed as either:*
  1. *Open-ended*
     - *Try to anticipate the response you will get*
     - *Well suited for getting opinions*
  2. *Closed*
     - *Use when all the options may be listed*
     - *When the options are mutually exclusive*

QUESTIONNAIRES LANGUAGE

- *Simple*
- *Specific*
- *Short*
- *Not patronizing*
- *Free of bias*
- *Addressed to those who are knowledgeable*
- *Technically accurate*
- *Appropriate for the reading level of the respondent*
- *Choose words carefully*

MEASUREMENT SCALES

NOMINAL SCALES

- *Nominal scales are used to classify things*
- *It is the weakest form of measurement*
- *Data may be calculated*

What type of software do you use the most?
(1) = Word Processor
2 = Spreadsheet
3 = Database
4 = An Email Program

INTERVAL SCALES

- *An interval scale is used when the intervals are equal*
- *There is no absolute zero*
- *Examples of interval scales include the Fahrenheit or Centigrade scale*

How useful is the support given by the Technical Support Group?

NOT USEFUL AT ALL				EXTREMELY USEFUL
1	2	3	(4)	5

PROBLEMS WITH SCALES

- *The rater (respondent) may do certain common errors in the answers.*
- *Major types of rater errors are:*

LENIENCY	CENTRAL TENDENCY	HALO EFFECT
○*Caused by easy raters* ○*Leniency error is the tendency of a manager to rate an employee higher than his actual performance.* ○*Reasons that a manager might do this could include avoiding confrontations, or feeling that by giving the employee a high rating, he will work harder to live up to the rating.*	○*Central tendency occurs when respondents rate everything as average* ○*Improve by making the differences smaller at the two ends* ○*Adjust the strength of the descriptors* ○*Create a scale with more points*	○*When the impression formed in one question carries into the next question.* ○*The halo effect is when a manger forms a positive impression of an employee's skill in one area and then gives her high ratings across all rating criteria* ○*Solution is to place one trait on each page of questionnaire*

DESIGINING THE QUESTIONNAIRES

- *Allow ample white space*
- *Allow ample space to write or type in responses*
- *Make it easy for respondents to clearly mark their answers*
- *Be consistent in style*
- *Order of Questions*
  1. *Place most important questions first*
  2. *Cluster items of similar content together*
  3. *Introduce less controversial questions first*

ADMINISTERING QUESTIONNAIRES

- *Administering questionnaires has two main questions:*

Who in the organization should receive the questionnaire?

How should the questionnaire be administered?

29

METHODS OF ADMINISTRERING THE QUESTIONNAIRES

1. *Convening all concerned respondents together at one time*
2. *Personally administering the questionnaire*
3. *Allowing respondents to self-administer the questionnaire*
4. *Mailing questionnaires*
5. *Administering over the Web or via email*

30

ADMINISTERING OVER WEB/EMAIL

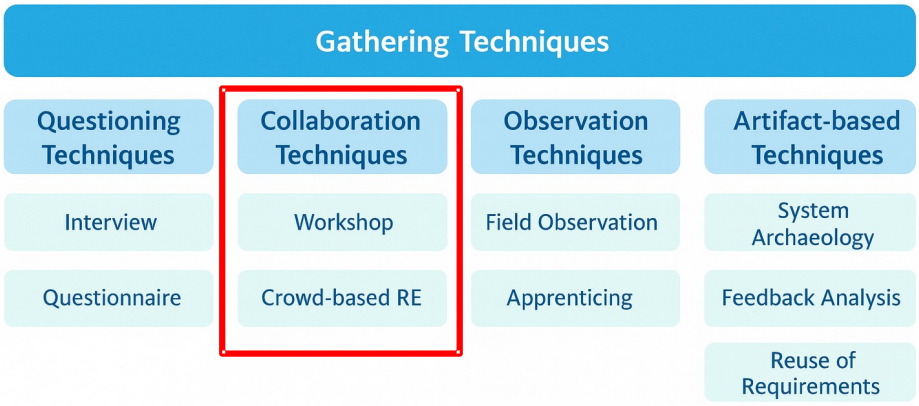
- *Electronically submitting questionnaire:*
  - *reduced costs*
  - *collecting and storing the results electronically*

Name	Appearance	Purpose
One-line text box		Used to obtain a small amount of text and limit the answer to a few words
Scrolling text box		Used to obtain one or more paragraphs of text
Check box		Used to obtain a yes-no answer (e.g., Do you wish to be included on the mailing list?)
Radio button		Used to obtain a yes-no or true-false answer
Drop-down menu		Used to obtain more consistent results (Respondent is able to choose the appropriate answer from a predetermined list [e.g., a list of state abbreviations])
Push button	Button	Most often used for an action (e.g., a respondent pushes a button marked "Submit" or "Clear")

Ways to Capture Responses When Designing a Web Survey

31

Gathering Techniques



32

Workshop

Workshop

- *A* **workshop** *is a broad term for group-oriented elicitation techniques.*
- *The size of workshops can vary:*
  - *From small, informal meetings*
  - *To structured events involving dozens or even hundreds of stakeholders*



Requirements Workshop

- *A structured session with a carefully selected group of stakeholders and subject matter experts.*
- *Participants collaborate to define, create, refine, and finalize deliverables (e.g., models, documents) that capture user requirements.*
- *The workshop is facilitated and moderated by the requirements engineer.*

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33

Workshop – Benefits and Uses

- *Provides broad insights in a relatively short time by leveraging active interaction among participants.*
- *For deeper details, follow-up techniques such as interviews or questionnaires may be applied.*
- *Workshops are also valuable in supporting* **creativity techniques** *to generate innovative solutions.*



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35

Crowd-based RE

- *Also known as* **platform-based Requirements Engineering**.
- *Turns elicitation into a participatory activity.*
- *A large group of stakeholders (especially users) provides feedback to achieve more accurate requirements.*
- *The strength of this approach lies in the* **diversity of knowledge, skills, and expertise** *among participants – often referred to as the "wisdom of the crowd".*



Crowd-based RE

Involves gathering requirements data from a large and diverse population via digital platforms (e.g., social media, online surveys, feedback portals, app stores, or dedicated crowdsourcing platforms like Amazon Mechanical Turk or IdeaScale).

STEP 1

Crowd Recruitment

- Identify and invite a large, diverse group of potential contributors (the "crowd")e role.
- Example: Inviting mobile app users to share improvement ideas

STEP 2

Requirements Elicitation

- Collect inputs such as ideas, feedback, bug reports, feature requests, and user stories
- Example: Gathering feature suggestions via an online form or forum.

STEP 3

Requirements Analysis & Filtering

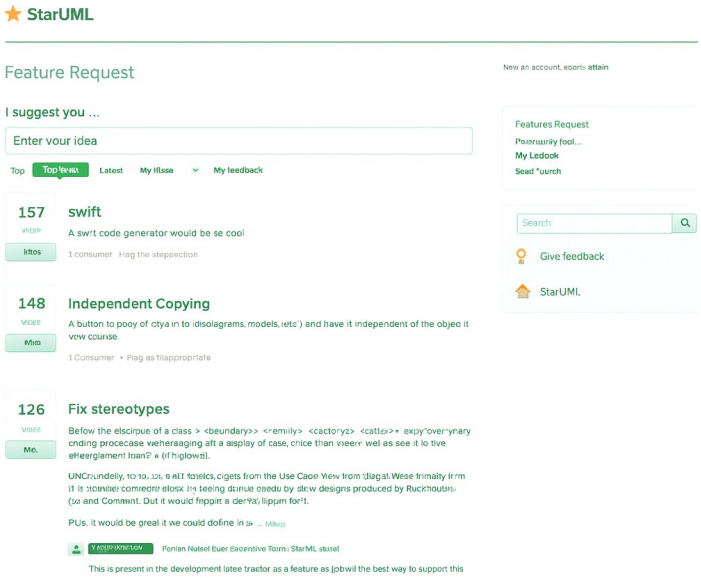
- AFilter, cluster, and prioritize useful requirements from the large volume of raw input
- Example: Using voting systems to identify the most requested features

STEP 4

Validation and Feedback

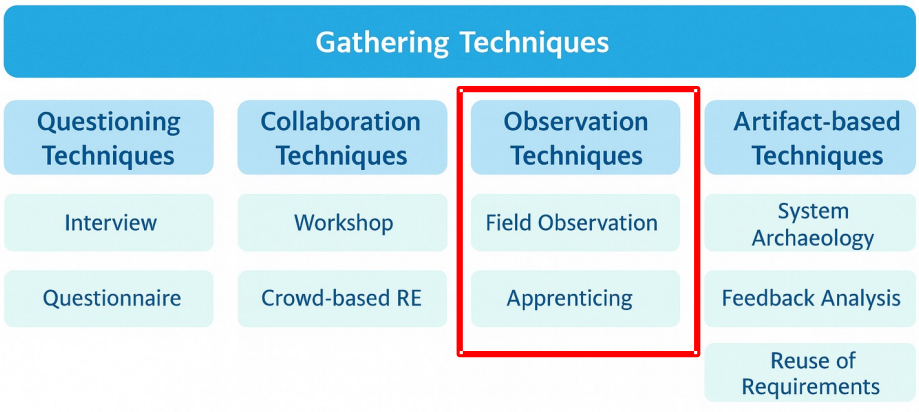
- Share the interpreted requirements or prototypes back to the crowd for validation and refinement
- Example: Beta testing or polls to confirm top features

Example – uservoice rating on new features



38

Gathering Techniques



39

Field Observation



- *The Requirements Engineer observes stakeholders in their usual work environment* **without interfering**.
- *Does not distract users.*
- *Does not interfere with the process itself.*
- *Requires careful preparation, keen observation skills, and a significant amount of time.*
- *Video recording can be very useful for capturing stakeholder behavior.*

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40

Field Observation

- *Helps uncover detailed requirements that stakeholders may not usually express, for example:*
- *Actions and behaviors that are too complex to describe in words.*
- *Routine activities, sequences, or data that are so common to stakeholders that they are often left unmentioned.*
- *Particularly effective for identifying* **dissatisfiers**.



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41

Apprenticing

- *Apprenticing is a* **participatory technique**.
- *Involves doing a short internship within the stakeholders' environment using existing systems.*
- *Experienced subject matter experts ("masters") guide the apprentice to gain deeper domain knowledge.*
- *The apprentice participates without interfering:*
  - *Takes on the role of a novice in the field.*
  - *Allowed to make mistakes and ask basic or "naive" questions.*
- **Goal**: *Develop a thorough understanding of the domain, business, and processes before beginning requirements elicitation.*

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43

Apprenticing

Executing the work of the user under their guidance

- Planning and preparation
- Experience in the real context, recording
- Deriving requirements



Strength ☺

- Challenging procedures and existing solutions
- Identifies basic factors
- Inversion of "guide-follower" relationship between RE and stakeholder

Weakness ☹

- Same as for field observation
- Time consuming for stakeholder
- Hurdle of needed domain knowledge for the requirements engineer

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Apprenticing

- *Follow-up interviews and questionnaires are often necessary to validate initial insights.*
- *Requirements gathered can then be documented and verified.*
- **Duration of apprenticeship** *varies depending on:*
  - *Process complexity, repetitiveness, and time availability of both master and apprentice.*
  - *Typically ranges from one day to several weeks.*
- *May be* **difficult or impractical** *to apply in certain domains (e.g., medicine, aviation, military).*

Gathering Techniques



Artifact-based Techniques

Gathering Techniques

- **Do not use stakeholders directly, but instead rely on work products such as documents, systems, images, audio, or video files as sources for requirements.**
- **Useful for uncovering satisfiers and dissatisfiers, sometimes in great detail.**
- **Particularly helpful when stakeholders are not easily available.**

Artifact-based Techniques

System Archaeology

Feedback Analysis

Reuse of Requirements

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System Archeology

- *Extract requirements from* **existing systems**, *such as:*
  - *Legacy systems*
  - *Competitor systems*
  - *Analogous systems*
- *Analyze their documentation (designs, manuals) or implementation (code, comments, scripts, user stories, test cases).*
- *Mainly applied when a legacy system is being replaced by a new system.*
  - *The new system must replicate the same functionality as the old one, or at least certain essential parts.*

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System Archeology

- *A* **time-consuming technique**.
- *Reveals detailed requirements and constraints that are not easily detected through other methods.*
- *Essential to confirm with stakeholders whether the identified requirements are still valid and relevant.*

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49

Feedback Analysis

- *Many ways to collect feedback from (potential) users and customers on an existing system or prototype:*
  - *Web surveys and contact forms*
  - *Beta or A/B testing*
  - *Social media*
  - *Customer remarks received via call centers*
- *Feedback data may be:*
  - **Structured** *(e.g., a 5-star rating in an app store)*
  - **Unstructured** *(e.g., written review comments)*
- *The volume of feedback is often very large, and analysis can be time-consuming.*

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50

Feedback Analysis

- *Feedback provides valuable insights into users'* **pains and gains**.
- *Negative scores and critical remarks help detect hidden* **dissatisfiers**.
- *Positive scores and compliments provide information about* **satisfiers**.
- *User comments may include innovative ideas that can be developed into* **delighters**.
- *Feedback analysis may result in:*
  - *Adjustments to existing requirements*
  - *Discovery of new requirements*

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Reuse of Requirements

- *Many organizations already have a* **large collection of requirements** *that were elicited and documented for previous systems.*
- *Many of these requirements may also be* **applicable to new systems**.
- *Especially useful for requirements derived from an* **overarching domain model**.
- *Reusing existing requirements can save significant* **time and money** *since elicitation can be skipped.*

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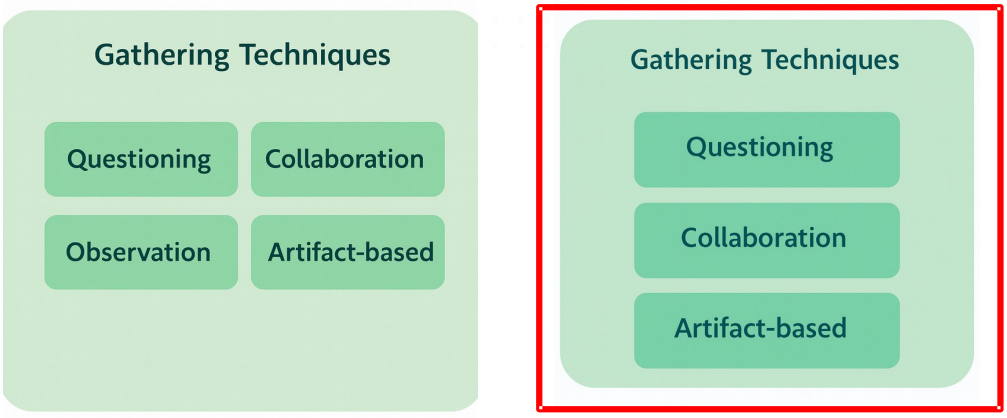
Reuse of Requirements

- *Reuse only works if the existing requirements are:*
  - *Up to date*
  - *Managed effectively*
  - *Easily accessible*
  - *Thoroughly documented*
- *Even if reuse is possible:*
  - *You must* **validate with stakeholders** *to confirm that these reusable requirements are still relevant and valid in the new context (or determine if adjustments are needed).*

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53

Design and Idea-generating Techniques



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54

Design and Idea-generating Techniques

- *In the past, Requirements Engineering (RE) focused on gathering and documenting requirements from stakeholders using traditional gathering techniques.*
- *Today, software is increasingly becoming a* **driver of innovation** *in many businesses.*
- *There is a growing need to reposition RE as a* **creative, problem-solving activity**.
- **Design and Idea-generating Techniques** *have emerge this need.*

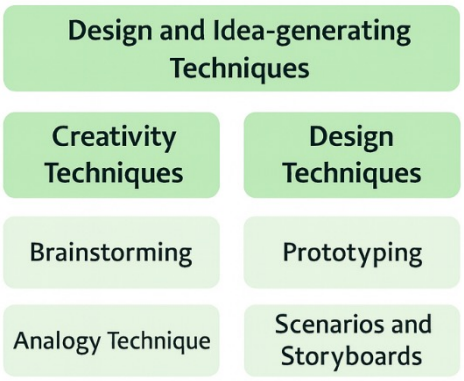


Design and Idea-generating Techniques

- *A broader concept in this area is known as* **design thinking**.
- *Different approaches include:*
- **d.school** *(https://dschool.stanford.edu)*
- **Designing for Growth** *(http://www.designingforgrowthbook.com)*
- *These approaches provide a wide range of techniques that can be applied to the* **elicitation of requirements**.



Design and Idea-generating Techniques



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57

Creativity Techniques

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58

Creativity Technique

- *Techniques encourage out-of-the-box and borderless thinking, as well as building upon others' ideas.*
- *Success is not guaranteed:*
  - *Several cognitive mechanisms in the brain must align to generate creative ideas.*
- *Preconditions for creativity:*
  - *Chance, Knowledge, Motivation, Safety/Security*
  - *Important to ensure these conditions are in place when applying creativity techniques*
  - *Must be met for participants to fully engage*

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59

Brainstorming

- *Encourages the development of new ideas for a specific question or problem.*
- *Important: Postpone judgment by separating idea generation from idea evaluation.*
- *General guidelines for brainstorming:*
  - *Prioritize quantity over quality.*
  - *Free association and visionary thinking are highly encouraged.*
  - *Combining and expanding on expressed ideas is allowed and welcomed.*
  - *Criticizing others' ideas is prohibited, even if the idea seems impractical.*
- *After brainstorming:*
  - *Generated ideas are categorized, assessed, and prioritized.*
  - *Selected ideas then serve as input for further requirements elicitation.*

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60

Analogy Technique

- *Supports the development of ideas for* **critical and complex topics**.
- *Uses* **analogies** *to stimulate thinking and generate new ideas.*
- *The outcome (success or failure) depends largely on selecting a* **proper analogy** *for the problem at hand.*
- *Compared to the original problem, the analogy can be:*
- **Close to**: *e.g., the same problem in another business.*
- **Distant from**: *e.g., comparing an organization with a living organism.*

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62

Design Architecture

- provide additional, explorative, or combinatorial techniques to elaborate ideas and gain further insights for a given idea
- rely heavily on visualiza-tion, team cooperation, and customer feedback

Design and Idea-generating Techniques

Design Techniques

Prototyping

Scenarios and Storyboards

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63

Prototype

- *Prototype: An intermediate work product created or released to generate feedback.*
- *Allows users to experiment with the system in a tangible way.*
- *Helps investigate unclear aspects during Requirements Engineering (RE) and before full implementation.*
- *Proper user guidance and feedback analysis are essential to derive new requirements.*
- *Particularly useful for identifying:*
  - *Non-functional requirements*
  - *Dissatisfiers and constraints*
  - *Characteristics difficult to define upfront in models or documentation*

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64

Prototype

- **Exploratory vs. Evolutionary Prototypes**
- *Levels of fidelity:* **Low, Medium, High**
- *Common types of exploratory prototypes:*
  - *Wireframes / Paper Prototypes*
  - *Mock-ups*
  - *Native Prototypes*

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65

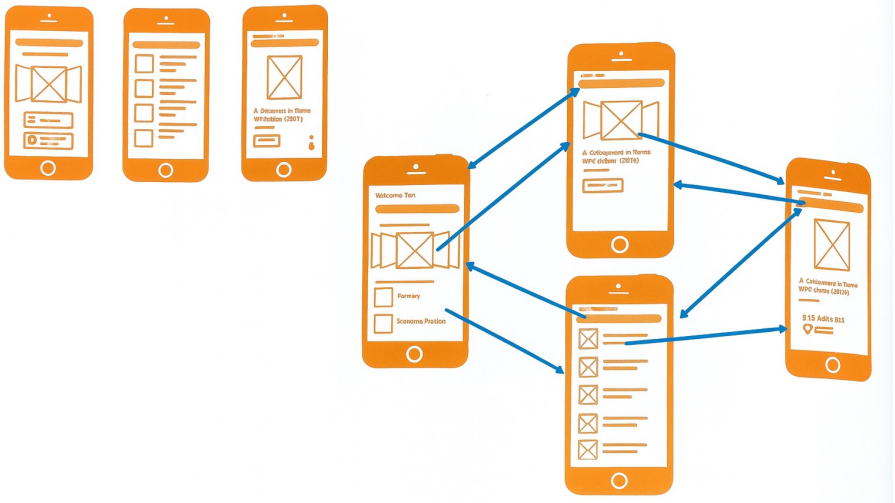
Prototype

- *Requirements Engineers mainly use exploratory prototypes as a tool for requirements elicitation and validation.*
- *In elicitation, prototypes act as a form of specification by example:*
  - *Useful when stakeholders cannot clearly articulate what they want.*
  - *Demonstrates what the final system might look like, helping stakeholders refine and shape their requirements.*
- *In validation, prototypes provide a powerful way to validate the adequacy of requirements.*

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66

Wireframes



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67

Scenarios and Storyboards

- *In RE, an ordered sequence of interactions between partners, in particular between a system and external actors*
- *May be a concrete sequence (instance scenario) or a set of potential sequences (type scenario, use case)*
- *Scenarios are documented in*
  - *written form (like story)*
  - *visual form (called a storyboard)*

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68

Scenarios and Storyboards

- *Storyboard: a series of panels that show sketches of a scene or action in a series of shots (as for a film, TV show or commercial)*
- *In Requirements Engineering:*
  - *Scenarios can be used to create an outline of the usage of a system*
  - *A storyboard can be used to describe a flow of actions for a system*

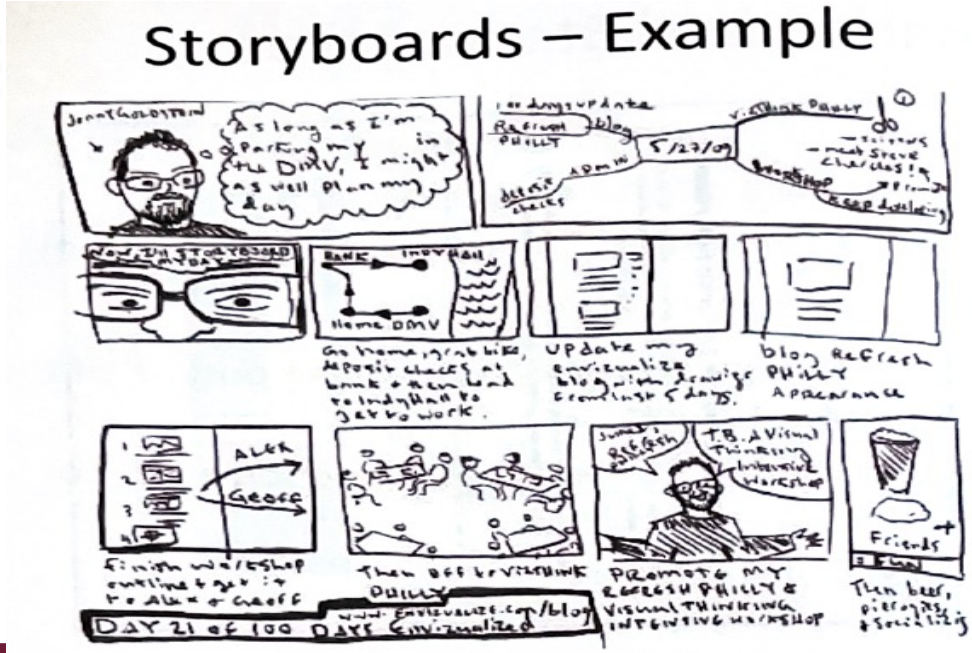
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69



70

Storyboards – Example



71

Storyboards – Example



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Innovating Solutions

72

THANK YOU

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79